

Reten Five Solution 5g/100ml

To be supplied only on the prescription of a registered medical practitioner

Each 100ml contains Minoxidil 5g

Product Description

Clear colourless solution.

PHARMACODYNAMICS

Minoxidil stimulates hair growth in alopecia androgenetica males and females when applied topically. The mechanism by which minoxidil stimulates hair growth and reverses the hair loss process is not fully established, but possible mechanisms may include the following:

- increasing the diameter of the hair shaft and stimulating of hair follicles,
- stimulating anagen (active) growth,
- prolonging the anagen phase
- stimulation of telogen phase (resting hair follicles) into anagen phase (active growth)
- enhancing microcirculation to hair follicles (as peripheral vasodilator).

The Vascular Endothelial Growth Factor (VEGF) is stimulated by minoxidil and VEGF is presumably responsible for the increased capillary fenestration, indicative of a high metabolic activity, observed during the anagen phase. Hair growth can be expected approximately after 4 or more months of application of use and is variable among patients. Upon discontinuation of treatment with minoxidil, new hair growth stops and pre-treatment appearance may return within 3 – 4 months.

PHARMACOKINETICS

Following topical application, minoxidil is poorly absorbed from normal intact skin, with an average of 1.7% (range 0.3 – 4.5%) of the total applied dose ultimately reaching the systemic circulation. Three major factors that can increase the absorption of minoxidil are increased of the dose applied, increased of the dosing frequency, and decreased of the barrier function of the stratum corneum.

Serum minoxidil level and systemic effects resulting from the administration of minoxidil are governed by the drug's absorption rate through the skin. Minoxidil is not bound to plasma protein and does not cross blood brain barrier. Minoxidil and its metabolites are haemodialysable, and excreted principally in the urine within 4 days.

INDICATION

Reten Five Solution 5g/100ml is indicated for the treatment of male pattern baldness for both male and female. Observations of patients suggest that the rate of balding progression may be slowed by the use of this product. Onset and degree of hair growth stimulation vary among patients. An inverse relationship may exist between duration of follicular dormancy and favourable response to treatment.

RECOMMENDED DOSE

Apply 6 sprays (1ml) of **Reten Five Solution 5g/100ml** at a time, 2 times a day to the areas of baldness or hair loss. Massage scalp gently with fingertips. The total daily dose should not exceed 12 sprays (2ml)

CONTRAINDICATION

Reten Five Solution 5g/100ml is contraindicated in patients with a history of hypersensitivity to minoxidil, alcohol and propylene glycol.

WARNINGS AND PRECAUTIONS

- Before using **Reten Five Solution 5g/100ml**, the user should determine that the scalp is normal and healthy. Topical minoxidil should not be applied to inflamed, infected, irritated or painful scalp skin.
- Topical minoxidil is only indicated for the treatment of alopecia androgenetica and should not be used in other types of hair loss for example when there is no family history of hair loss, hair loss is sudden and/or patchy, hair loss is due to childbirth, or the reason for hair loss is unknown.
- The patient should stop using **Reten Five Solution 5g/100ml** and see a doctor if hypotension is detected or if the patient is experiencing chest pain, rapid heartbeat, faintness or dizziness, sudden unexplained weight gain,

swollen hands or feet or persistent redness or irritation of the scalp or other unexpected new symptoms occur.

- Patients with known cardiovascular disease or cardiac arrhythmia should contact a physician before using **Reten Five Solution 5g/100ml**.
- Some patients have experienced changes in hair colour and/or texture with use of **Reten Five Solution 5g/100ml**.
- **Reten Five Solution 5g/100ml** is for external use only. Do not apply to areas of the body other than the scalp.
- Using more than the recommended dose or more often will not improve results.
- Unwanted hair growth may be caused by the transfer of the product to areas other than the scalp.
- Hands should be washed thoroughly after applying the solution. Inhalation of the spray mist should be avoided.
- Some consumers reported increased hair shedding upon initiation of therapy with **Reten Five Solution 5g/100ml**. This is most likely due to minoxidil's action of shifting hairs from the resting telogen phase to the growing anagen phase (old hairs fall out as new hairs grow in their place). This temporary increase in hair shedding generally occurs two to six weeks after beginning treatment and subsides within a couple of weeks. If shedding persist (>2weeks), users should stop using **Reten Five Solution 5g/100ml** and consult their doctor.
- Accidental ingestion may cause serious cardiac adverse events. Therefore this product has to be kept out of the reach of children.
- **Reten Five Solution 5g/100ml** contains alcohol, which will cause burning and irritation of the eye. In the event of accidental contact with sensitive surfaces (eye, abraded skin and mucous membranes) the area should be bathed with large amounts of cool tap water.
- **Reten Five Solution 5g/100ml** contains propylene glycol, which may cause skin irritation.

DRUG INTERACTION

- This product should not be used concomitantly with other medications applied topically on the scalp.
- Topical drugs, such as corticosteroids, tretinoin, dithranol or petrolatum, which alter the stratum corneum barrier, could result in increased absorption of minoxidil if applied concurrently. Although it has not been demonstrated clinically, there exists the theoretical possibility of absorbed minoxidil potentiating orthostatic hypotension caused by peripheral vasodilators.
- Guanethidine has been reported to interact with oral formulations of minoxidil resulting in rapid and pronounced lowering of blood pressure. There is a theoretical possibility that topical minoxidil may also interact with guanethidine.

PREGNANCY AND LACTATION

Reten Five Solution 5g/100ml should not be used during pregnancy or lactation.

SIDE EFFECTS

System Organ Class	Frequency	Preferred term
Immune system disorders	Common	Hypersensitivity reactions (including face oedema, generalised erythema, pruritus generalised, swelling face and throat tightness)
	Not known	Angioedema (including lip oedema, lip swelling, oedema mouth, oropharyngeal swelling, pharyngeal oedema, swollen tongue and tongue oedema)
Psychiatric disorders	Not known	Depressed mood
Nervous system disorders	Very common	Headache
	Uncommon	Dizziness
Eye disorders	Not known	Eye irritation
Cardiac disorders	Common	Chest pain
	Uncommon	Palpitations
	Not known	Heart rate increased
Vascular disorders	Not known	Hypotension
Respiratory, thoracic and mediastinal disorders	Uncommon	Dyspnoea
Gastrointestinal disorders	Uncommon	Nausea
	Not known	Vomiting
Skin and subcutaneous tissue disorders	Common	Hypertrichosis (unwanted non-scalp hair including facial hair growth in women), pruritus (including rash pruritic generalised and eye pruritus), rash (including pustular, popular, generalised, vestibular

		and macular rash), dermatitis (including contact, allergic, atopic and seborrheic dermatitis)
	Rare	Changes in hair texture
	Not known	Dry skin, skin exfoliation (including exfoliative rash, dermatitis, exfoliative), acne (acneiform rash), temporary hair loss, changes in hair colour
General disorders and administration site conditions	Common	Oedema peripheral
	Not known	Application site reactions (these sometimes involve nearby structures like the ears, face and typically consist of pruritus, irritation, pain, rash, oedema, dry skin, erythema, burning sensation and rash erythematous but can sometimes be more severe and include exfoliation, dermatitis, blistering, bleeding and ulceration)
Investigations	Common	Weight increased

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

This product may cause dizziness or hypotension. If affected, patients should not drive or operate machinery.

SYMPTOMS AND TREATMENT OF OVERDOSE

Increased systemic absorption of minoxidil may potentially occur if higher-than-recommended doses of **Reten Five Solution 5g/100ml** are applied to larger surface areas of the body or areas other than the scalp.

Because of the concentration of minoxidil in **Reten Five Solution 5g/100ml**, accidental ingestion has the potential of producing systemic effects related to the pharmacological action of the drug (2ml of **Reten Five Solution 5g/100ml** contains 100mg minoxidil; the maximum recommended adult dose for oral minoxidil administration in the treatment of hypertension). Signs and symptoms of minoxidil overdose would primarily be cardiovascular effects associated with sodium and water retention. Tachycardia, hypotension, dizziness and lethargy can also occur.

Treatment

The treatment of minoxidil overdosage should be symptomatic and supportive.

Fluid retention can be managed with appropriate diuretic therapy. Clinically significant tachycardia can be controlled by administration of a beta-adrenergic blocking agent.

INSTRUCTIONS FOR USE

For external use only.

Do not apply to other parts of the body other than scalp.

Apply 6 sprays (1ml) of Reten Five Solution 5g/100ml per application, twice daily to the area of hair loss. Massage scalp gently with fingertips. The total daily dose should not exceed 12 sprays (2ml). Wash hands thoroughly after application.

EXCIPIENTS

Propylene glycol, alcohol 95%, purified water, sodium thiosulphate, phosphoric acid.

Storage Condition

Store below 30°C.

Keep in tight container.

Protect from light.

Keep out of the reach of children.

Jauhi daripada kanak-kanak.

Pack Size

Bottle of 30ml and 60ml

Shelf life: Please refer to the outer box label.

Manufactured by:

T.O. Pharma Co. Ltd.

101 Soi 124, Ladprao Rd. Bangkok 10310, Thailand.

Revision Date: 26 January 2022